

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 3 Winter 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AF1191PC303 - SURVEYING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 The main principle of surveying is to
 - A) Work from details to whole
 - B) Work from whole to part
 - C) Work from centre to boundary
 - D) Work from boundary to centre
- 2 Which correction is always negative in chain surveying?
 - A) Temperature correction
 - B) Pull correction
 - C) Sag correction
 - D) Slope correction
- 3 A well-conditioned triangle should have no angle less than
 - A) 15°
 - B) 30°
 - C) 45°
 - D) 60°
- 4 The angle between magnetic meridian and true meridian is called
 - A) Local attraction
 - B) Dip
 - C) Magnetic declination
 - D) Bearing
- 5 Which bearing system uses values from 0° to 360° ?
 - A) Quadrantal bearing system
 - B) Reduced bearing system
 - C) Whole circle bearing system
 - D) Arbitrary bearing system
- 6 Which bearing system uses values from 0° to 360° ?
 - A) Quadrantal bearing system
 - B) Reduced bearing system
 - C) Whole circle bearing system
 - D) Arbitrary bearing system

- 7 Local attraction is detected when
 - A) $FB \neq BB \pm 180^\circ$
 - B) Declination is zero
 - C) Dip is maximum
 - D) Bearings are reduced
- 8 A level surface is
 - A) Plane surface
 - B) Curved surface parallel to earth
 - C) Horizontal surface
 - D) Vertical surface
- 9 Temporary adjustment of a theodolite includes
 - A) Axis adjustments
 - B) Plate level adjustment
 - C) Centering, levelling and focusing
 - D) Collimation adjustment
- 10 Closing error in a traverse occurs due to
 - A) Instrumental errors
 - B) Observational errors
 - C) Computational errors
 - D) All of the above
- 11 The primary purpose of reconnaissance survey is to
 - A) Prepare detailed drawings
 - B) Estimate quantities
 - C) Select the best possible route
 - D) Set out works
- 12 Setting out of works refers to
 - A) Preparation of drawings
 - B) Transferring plan details to ground
 - C) Cost estimation
 - D) Soil investigation

Q2. Solve the following.

- A) Explain the fundamental principles of chain surveying and discuss the importance of "working from whole to part". <https://www.batuonline.com>
- B) The area of the plan of an old survey plotted to a scale of 10 metres to 1 cm measures 6 now as 100.2 sq. cm as found by a planimeter. The plan is found to have shrunk so that a line originally 10 cm long now measures 9.7 cm only. Find (i) the shrunk scale, (ii) true area of the survey.

Q3. Solve the following.

- A) Differentiate between the Prismatic Compass and the Surveyor's Compass by explaining their reading arrangements, bearing systems, mode of observation, and suitability for surveying work.

- B) The following bearings were taken in running a closed compass traverse while 6 surveying in Mumbai:

Line	F.B.	B.B.
AB	48°25'	230°00'
BC	177°45'	356°00'
CD	104°15'	284°55'
DE	165°15'	345°15'
EA	259°30'	79°00'

- (i) State the stations which are affected by local attraction and by how much.
(ii) Determine the correct bearings.
(iii) Calculate the true bearings if the declination was 1°30' W.

Q4. Solve Any Two of the following.

- A) Define "level line," "horizontal line," and "datum" in the context of levelling with 6 proper diagram.
- B) Following is the page of a level field book. The readings in the level book were 6 written with pencil and some of these got erased. The erased readings are marked with question marks. Calculate the missing readings.

Station	B.S.	I.S.	F.S.	Rise	Fall	R.L.	Remarks
1	?					150.00	B.M.
2		2.457			0.827	?	
3		2.400		0.057		?	
4	2.697		?		?	148.070	C.P.
5	?		2.051	0.646		148.716	
6		2.500		1.068		149.784	C.P.
7		2.896			?	149.388	
8		?			0.124	?	
9			2.672	0.348		149.612	

- C) Define "contour" and explain the main "characteristics of contours." How are contour 6 maps useful in engineering and planning applications?

Q5. Solve Any Two of the following.

- A) Describe the main components of a vernier transit theodolite. Explain the "temporary 6 adjustments" that must be performed before taking any observation with the instrument.
- B) Explain the concept of "traversing" using a theodolite. Discuss the terms "closing 6 error" and "distribution of closing error" in the context of a closed traverse.
- C) Explain the difference between "vernier theodolite" and "micro-optic theodolite." 6

Q6. Solve Any Two of the following.

- A) Explain: i) Reconnaissance, ii) Preliminary Survey, and iii) Location Survey. 6
- B) What is "setting out works" in engineering surveying? 6
- C) What is tunnel correlation? Explain the methods used for connecting underground and 6 surface surveys.